

(g) Acids, bases and salt preparations	
Students should:	
2.34	know the general rules for predicting the solubility of ionic compounds in water: • common sodium, potassium and ammonium compounds are soluble • all nitrates are soluble • common chlorides are soluble, except those of silver and lead(II) • common sulfates are soluble, except for those of barium, calcium and lead(II) • common carbonates are insoluble, except for those of sodium, potassium and ammonium • common hydroxides are insoluble except for those of sodium, potassium and calcium (calcium hydroxide is slightly soluble).
2.35	understand acids and bases in terms of proton transfer
2.36	understand that an acid is a proton donor and a base is a proton acceptor
2.37	describe the reactions of hydrochloric acid, sulfuric acid and nitric acid with metals, bases and metal carbonates (excluding the reactions between nitric acid and metals) to form salts
2.38	know that metal oxides, metal hydroxides and ammonia can act as bases, and that alkalis are bases that are soluble in water
2.39	describe an experiment to prepare a pure, dry sample of a soluble salt, starting from an insoluble reactant
2.40C describe an experiment to prepare a pure, dry sample of a soluble salt, starting from an acid and alkali	
2.41C describe an experiment to prepare a pure, dry sample of an insoluble salt, starting from two soluble reactants	
2.42	<i>practical: prepare a sample of pure, dry hydrated copper(II) sulfate crystals starting from copper(II) oxide</i>
2.43C practical: prepare a sample of pure, dry lead(II) sulfate	